

Intro to Electronics Task

Task1:

1. Identifying the needs	Voltage and current each part needs?	Battery voltage higher or lower than what's needed?	Need protection against connecting the battery backwards?
Load A	Needs 9V DC . Standard coil current is approximately 40 mA to 50 mA	The battery voltage is higher than the 9V required by Load A (12.6V > 9V) It must be stepped down.	No
			No

Load B	Needs 5V DC and an operating current of 15 mA	The battery voltage (12.6V) is higher than the 5V required by Load B , must be stepped down	
3S Li-ion battery	Voltage: 12.6V Sized to provide total load current	The battery provides 12.6V peak, which is higher than all subsystem operating voltages	Yes Connecting the battery backwards will reverse-bias all regulators and ICs, leading to component damage and battery fire hazards.

2. Choose the components:

1. For Load A (9V)

- **Regulator :** LDO , LM7809
- **Why it was picked:** The difference between the 12.6V battery and 9V output is only 3.6V, and the relay draws a small current about 45 mA . The total thermal dissipation is negligible , so an LDO provides a clean, simple, and low-cost solution without needing bulky switching inductors.

2. For Load B (5V HC-SR04 Sensor)

- **Regulator Choice:** Buck Converter (Switching Regulator, LM2596-5.0).
- **Why it was picked:** Stepping down from 12.6V to 5V involves a large 7.6V drop. A Buck converter steps this down with high electrical efficiency of more than 85%, preventing unnecessary battery drains and avoiding heat generation inside.

3. Protection Circuit

- **Protection Choice:** P-Channel MOSFET circuit (High-Side reverse polarity switch).
- **Why it was picked:** Unlike a standard series rectifier diode that wastes power and causes a 0.7V drop across the battery line, a P-Channel MOSFET acts as an ideal

electronic switch with virtually zero resistance when forward-biased, while completely isolating and protecting the circuitry if the battery is reversed.

3. Check the datasheet

1. Load A Regulator: LM7809 (9V Linear Regulator)

- Datasheet Confirmation:
 - Input Voltage Range: 11.5V to 35V (Dropout voltage: 2.0V). Fully covers the 12.6V Li-ion battery peak.
 - Output Current: Rated up to 1.5A, which easily handles the 40 mA to 50 mA relay coil load.
- Why it fits : The LM7809 input range safely tolerates the 12.6V battery voltage without risking overvoltage breakdown. Its 1.5A output rating provides abundancy over the relay's tiny 50mA current demand. Since power loss is only 0.16W, it delivers a stable 9V rail with minimal external passive components and no heat sink required.

2. Load B Regulator: LM2596-5.0 (5V Buck Switching Regulator)

- Datasheet Confirmation:
 - Input Voltage Range: 4.5V to 40V. Fully covers the 12.6V battery voltage.
 - Output Current: Up to 3.0A continuous load current, easily exceeding the 15 mA sensor current.
- Why it fits : The LM2596-5.0 features a wide 40V input limit that effortlessly handles the 12.6V battery supply and absorbs

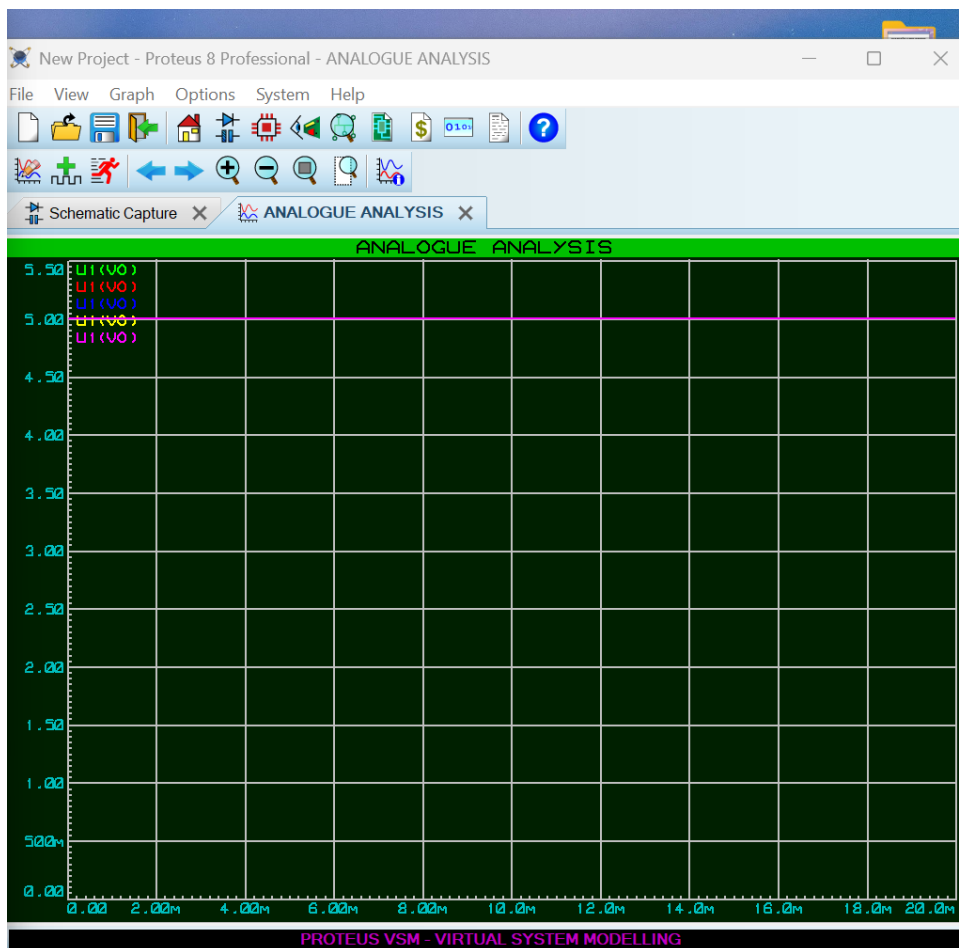
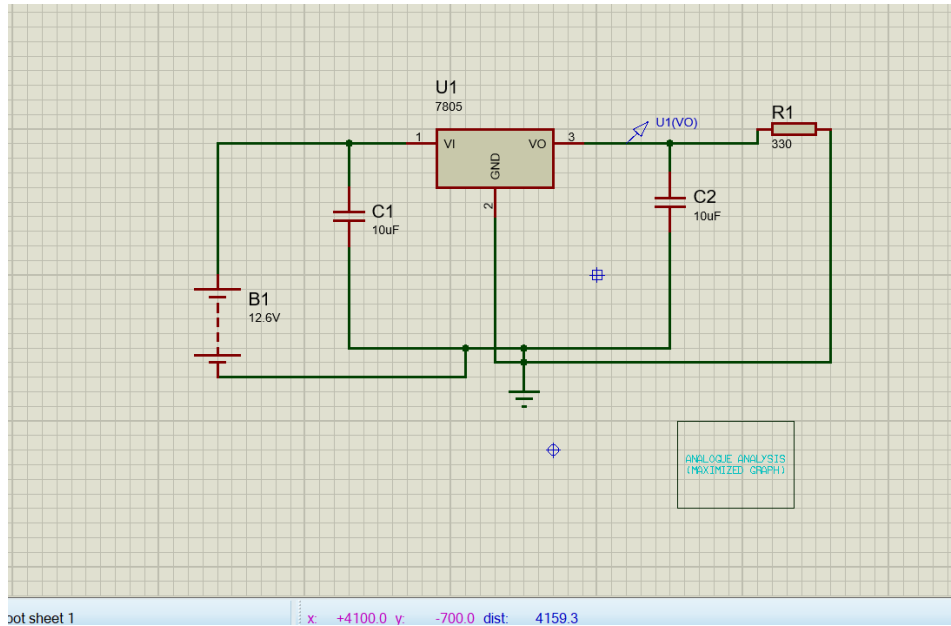
voltage transients. It delivers high step-down power conversion efficiency, eliminating unnecessary battery power waste across the large 7.6V drop. Its 3A rating leaves massive current capacity to power additional 5V digital sensors later.

3. Protection Component: AO3401A (P-Channel MOSFET)

- Datasheet Confirmation:
 - Input/Maximum Voltage: $V_{DS} = -30V$, easily blocking full -12.6V reverse polarity faults.
 - Continuous Current: $I_D = -4.0A$, more than sufficient for the total current.
- Why it fits : The AO3401A is rated for -30V, safely withstanding the reverse voltage if the 12.6V battery is plugged in backwards. Its low internal resistance (R_{DS_on} approx 50 mOhm) prevents unwanted voltage drop and power loss during regular forward operation. It delivers complete reverse-bias protection without the 0.7V diode drop that would degrade battery efficiency.

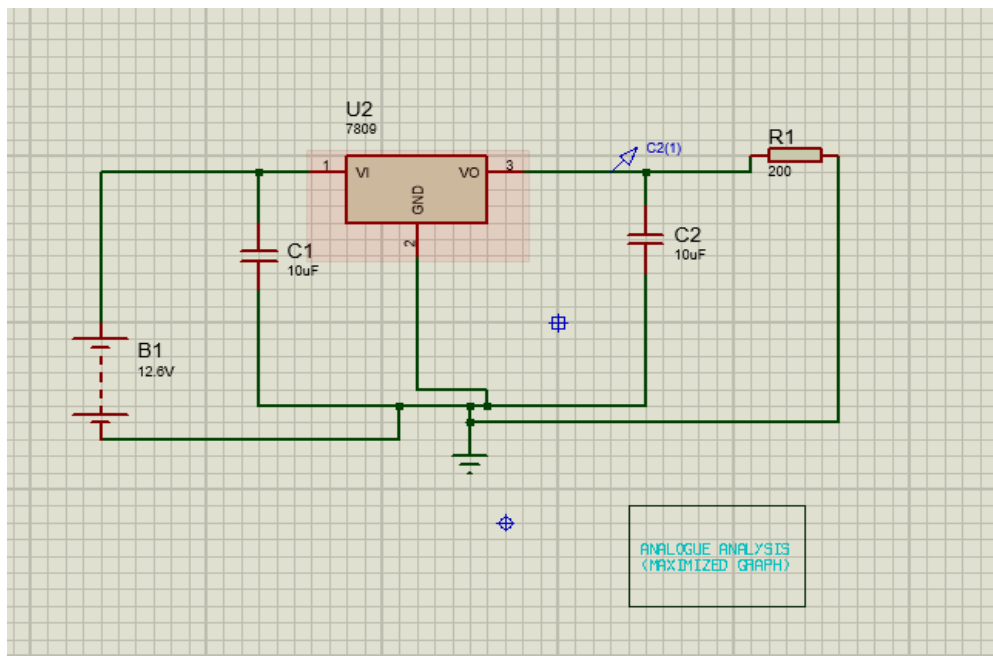
4. Regulator Circuit:

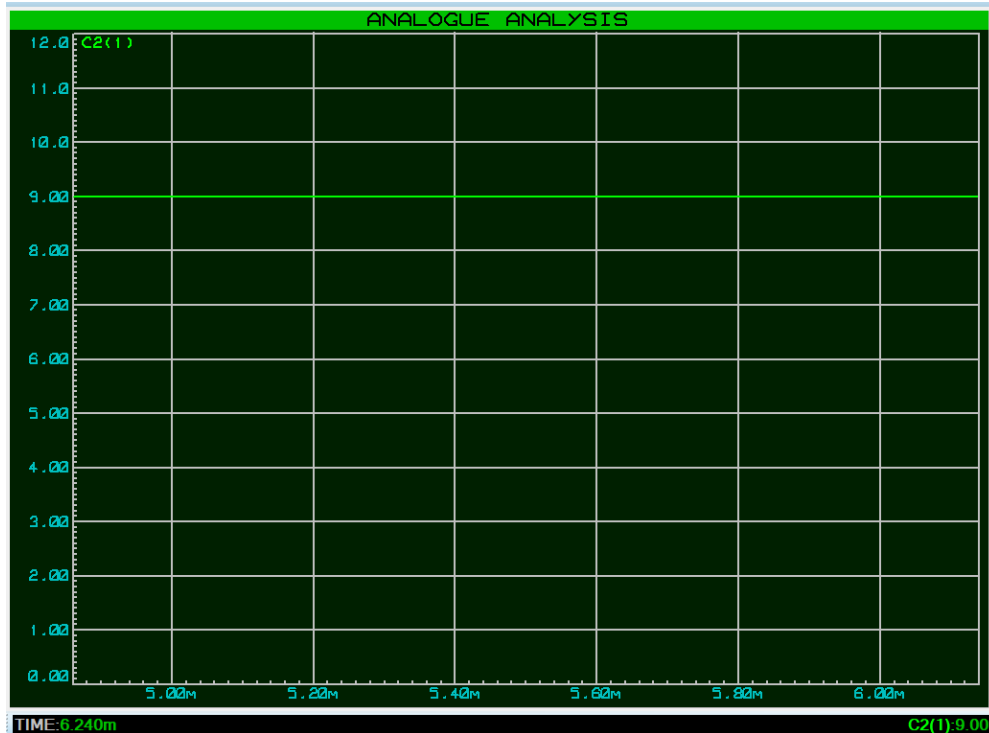
For Load B:



The simulation verifies the 5V voltage regulator circuit designed for Load B (HC-SR04 Ultrasonic Sensor). The 7805 regulator steps down the 12.6V battery input to 5.0V output. The sensor is modeled using an equivalent 330 Ohm load resistor, which draws the required 15 mA operating current to confirm voltage stability under real load conditions.

For Load A:





This simulation verifies the 9V voltage regulator circuit designed for Load A (Re22-Relay). The LM7809 regulator steps down the 12.6V battery supply to a stable 9.0V output rail. The relay coil is modeled using an equivalent 200 Ohm load resistor, which draws the required 45 mA operating current to verify that the output remains steady under load

Task 2:

1. Resistors (Designators: R1, R2...)

On the board, surface-mount resistors look like tiny black rectangular chips with silver end caps, while through-hole resistors look like small cylindrical bodies with colored stripes.

- **Function in the Circuit:** Resistors control and limit the amount of electrical current flowing through different parts of

the board, and they create precise voltage levels when used in pairs as voltage dividers.

- **Why It Is Needed:** Without current-limiting resistors, sensitive parts like indicator LEDs or control inputs would draw too much current and burn out. In adjustable voltage regulators, precision resistor dividers sample the output voltage and feed it back into the regulator chip so it knows exactly when to adjust power delivery to keep the output steady.

2. Capacitors (Designators: C1, C2...)

two main physical forms on power boards: large cylindrical or can-shaped components (electrolytic capacitors) and tiny rectangular brown or yellowish chips (ceramic capacitors).

- **Function in the Circuit:** Capacitors store electrical energy in an electric field and filter unwanted noise or voltage ripples out of the power lines.
- **Why It Is Needed:** When motors, relays, or digital microcontrollers turn on, they draw sudden surges of current that can cause the battery voltage to dip. Large bulk capacitors discharge instantly to support the power rail and keep the voltage stable. At the same time, smaller ceramic bypass capacitors placed directly next to the regulator IC pins absorb high-frequency electrical noise and divert it harmlessly to Ground before it can interfere with sensitive sensor readings.

3. Voltage Regulators and ICs (Designators: U1, U2...)

They are typically black packaged chips with three or more metal pins and often include a metal tab for heat dissipation.

- **Function in the Circuit:** The voltage regulator acts as the central control element of the power supply, stepping down the higher, unregulated 12.6V battery voltage to fixed, reliable operating levels like 9V or 5V.
- **Why It Is Needed:** Electronic components like sensors and microcontrollers will get damaged or fail to function if connected directly to the high, fluctuating voltage of a raw battery pack. The regulator ensures that regardless of whether the battery is fully charged (12.6V) or running low, the output voltage supplied to downstream devices remains locked at a safe, constant value.

4. Inductors (Designator: L1)

they usually look like thick black blocks or wire-wound coils wrapped around a core.

- **Function in the Circuit:** Inductors store electrical energy in a magnetic field whenever current passes through them and resist rapid changes in current flow.
- **Why It Is Needed:** Inductors are essential components in switching buck converter circuits (like the LM2596). They work alongside the high-speed switching chip by taking rapid, high-voltage pulses of power and smoothing them into a continuous, lower-voltage DC stream, allowing the circuit to step down voltage with very high efficiency and minimal heat loss.

5. Diodes (Designators: D1, D2...)

They are usually black rectangular components marked with a distinct white or silver stripe at one end (indicating the cathode pin).

- **Function in the Circuit:** A diode is a one-way valve for electricity that allows current to flow easily in only one direction while blocking it in reverse.
- **Why It Is Needed:** In switching power supplies, a fast-acting Schottky diode provides a safe return path for the inductor's current when the regulator's internal switch turns off. In general power inputs, diodes are also used to block reverse current if a battery is plugged in backward, preventing short circuits.

6. Connectors (Designators: J1, J2, P1...)

They are the physical plastic headers, screw terminals, or heavy-duty battery plugs soldered to the board edges.

- **Function in the Circuit:** Connectors provide secure physical and mechanical entry and exit points for incoming battery power and outgoing regulated voltage rails.
- **Why It Is Needed:** Connectors allow external hardware like the 3S Li-ion battery pack, the HC-SR04 ultrasonic sensor, and the relay module to be easily plugged in, removed, or serviced without having to solder wires directly to the board traces every time.